DIRECTORATE GENERAL OF VOCATIONAL AND TECNICAL EDUCATION



18th INTERNATIONAL MEB ROBOT COMPETITION

AUTONOMOUS VEHICLE CATEGORY GUIDE



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AUTONOMOUS VEHICLE CATEGORY COMPETITION RULES

1. GENERAL INFORMATION ABOUT THE COMPETITION

1.1. Objective

The Autonomous Vehicle Category aims to enable participants to apply their theoretical knowledge in the fields of image processing, autonomous navigation, and mechatronic system design in a practical platform. The competition focuses on the high-accuracy detection of environmental data, the real-time processing of this data, and its effective integration into control algorithms.

In this context, the optimisation of object recognition and lane tracking processes using image processing algorithms, feedback mechanisms used to ensure stability in motor driver systems, and the development of autonomous architectures that operate reliably in dynamic environments are at the forefront. The competition not only provides participants with a competitive environment but also offers a high-quality educational environment that prepares them for the autonomous and intelligent system technologies of the future.

The knowledge and skills acquired in this category extend beyond the competition process, making significant contributions to academic and industrial work in the fields of robotics, artificial intelligence, and autonomous systems. Participants can thus support their individual development, guide their careers, and add value to the future of technology.

The target audience for this competition consists of high school and university students, young engineering candidates, and technology enthusiasts who are interested in robotics, artificial intelligence, and autonomous systems and who want to combine their theoretical knowledge with image processing and control algorithms to produce innovative solutions to real-world problems.

1.2. Theme

The theme of the Autonomous Vehicle Category is based on demonstrating **autonomous decision-making and safe driving** skills on a track representing real traffic scenarios **using only camera-based image processing**. This category aims to enable participants to apply their theoretical knowledge in the fields of image reading, environmental perception, autonomous navigation, and control algorithms **in a practical environment.**

The tasks designed within the scope of the category are based on elements that correspond to real life, such as traffic lights, pedestrian crossings, level crossings, speed bumps, overtaking, dead-end streets, and parking. Thus, the competitors' competencies in **real-time data processing**, **correct decision-making**, and **developing stable autonomous systems in dynamic environments** are measured.

The Autonomous Vehicle Category is based on the principles of **technology-focused thinking**, **innovative solution development**, and **the holistic use of engineering skills**, in line with the general competition theme outlined in the application guide.

1.3. Participation Requirements and Team Structure

Students enrolled in secondary schools and universities may participate in the Autonomous Vehicle Category. Participation in the competition is based on **a team structure**.

Each team consists of **a maximum of 3 students**.

- For teams participating **at the secondary school level**, **the presence of a supervising teacher is mandatory**.
- **There is no requirement** for a mentor for teams participating at **the university level**.

The advisor is responsible for guiding the team throughout the competition process and **cannot take an active role** in the competition area.

During the competition, **no more than two** team members may be present **in the competition area** at any one time. In the event of a breach of this rule, the referees have the authority to stop the competition and apply the necessary sanctions.

1.4. The Critical Importance of Reading the Guidelines

The International MEB Robot Competition is a comprehensive and multidimensional competition that brings together students' technical knowledge, engineering skills and problem-solving abilities. The Autonomous Vehicle category expects competing teams to develop autonomous systems that demonstrate not only hardware adequacy but also the ability to perceive, make decisions, and navigate correctly in conditions similar to real traffic scenarios.

However, success in this competition is not limited to the mechanical structure of the vehicle or the complexity of its software. The true success of the competition lies in the technical competence of the autonomous vehicle and in **carefully reading and correctly interpreting the guidelines containing the competition rules, task descriptions and evaluation criteria at.**

The Application Guide and Autonomous Vehicle Category Guide are integral parts of the competition. These guides clearly define the entire technical and administrative framework of the competition, including the track layout, conditions for performing tasks, scoring system, penalties, and disqualification scenarios. Failure to comply with the rules specified in the guidelines may result in **point loss or disqualification**, even if the autonomous vehicle is technically successful.

Therefore, competing teams must:

- Read **the Application Guide** and **Category Guide** thoroughly before the competition,
- **Regularly follow the announcements and information** published on **robot.meb.gov.**tr throughout the competition process,
- Act in accordance with the current rules and regulations specified in the guides

are required.

Reading and understanding the guides correctly is as important an engineering process as designing an autonomous vehicle. The attention and diligence shown in this process constitutes the first and most important step towards success by preventing errors and loss of rights that may be encountered during the competition process.

All teams applying **to the Autonomous Vehicle Category of the 18th International MEB Robot Competition must read the Application Guide** (accessible from the **"Organisation"** menu at https://robot.meb.gov.tr), **which** contains the competition applications and general rules for the category.

2. TECHNICAL SPECIFICATIONS AND CONSTRAINTS OF THE ROBOT

2.1. Dimension and Weight Restrictions

Vehicles competing in the Autonomous Vehicle Category **must fully comply with the** following **dimensions, hardware, and software criteria** to pass the technical inspection and be accepted into the competition.

2.1.1.Dimensions:

Autonomous vehicles must fit **comfortably and without difficulty** into a box measuring **20 × 30 cm** during technical inspection.

2.1.2.Height:

The height of the autonomous vehicle must **not exceed 25 cm**.

2.1.3.Wheel Diameter:

The diameter of the wheels used on the vehicles **must not exceed 10 cm**.

2.1.4.Weight:

There are no **minimum or maximum weight limits** for autonomous vehicles.

2.1.5.Disqualification Condition:

Autonomous vehicles that violate **the** specified **size or height limits will not be accepted into the competition**.

2.1.6.Imaging System:

As the competition is based on **image processing**, **only cameras** are permitted in autonomous vehicles. There is no restriction on the number of cameras that may be used.

2.1.7.Sensor Ban:

Outside the camera; **the use of any electronic or mechanical sensors such as distance sensors, Lidar, IR sensors, ultrasonic sensors**, and similar **devices is prohibited**. Autonomous vehicles found to be using such sensors **will be disqualified**.

2.1.8.Control Board and Motors:

There are no restrictions on **the controller board (microprocessor/microcontroller)** to be used, the number of motors, or the rotational speed (**RPM**) of the motors.

2.1.9.Original Software:

The control algorithms of autonomous vehicles must be run using **software developed entirely by the competing team**. Ready-made, commercial or third-party control software may not be used.

2.1.10. Inspection Process:

During the technical inspections to be carried out prior to the competition, it will be verified that the autonomous vehicle **does not exceed the dimensions of 20 × 30 × 25 cm**, **does not contain any sensors other than a camera**, and complies with all conditions specified in the guidelines. Any non-compliance identified during the technical inspection may result in the autonomous vehicle **not being accepted for the competition**.

2.2. Permitted Materials and Components

In the Autonomous Vehicle Category, the use **of commercial software offering block-based or ready-made control solutions** (e.g. LEGO Spike standard software, etc.) **is prohibited**. The control and decision-making processes of autonomous vehicles must be carried out using **software developed by the competing team**.

2.3. Software and Control Requirements

Remote access modules such as infrared, Bluetooth, Wi-Fi, or Radio Frequency (RF) on the control board must be completely turned off during the competition.

If it is determined that the vehicle is being remotely controlled or that these modules are active during the competition, the competitor will be immediately disqualified.

2.4. Detailed Description of the Robot

Robots competing in the Autonomous Vehicle Category must be land-based autonomous vehicles designed to perform the tasks defined in the competition course. All mechanical and electronic components of the robot must be designed to fit within the dimensions (20x30x25) specified in the guide.

3. COMPETITION AREA AND TASKS

3.1. Shape and Dimensions of the Competition Area/Track

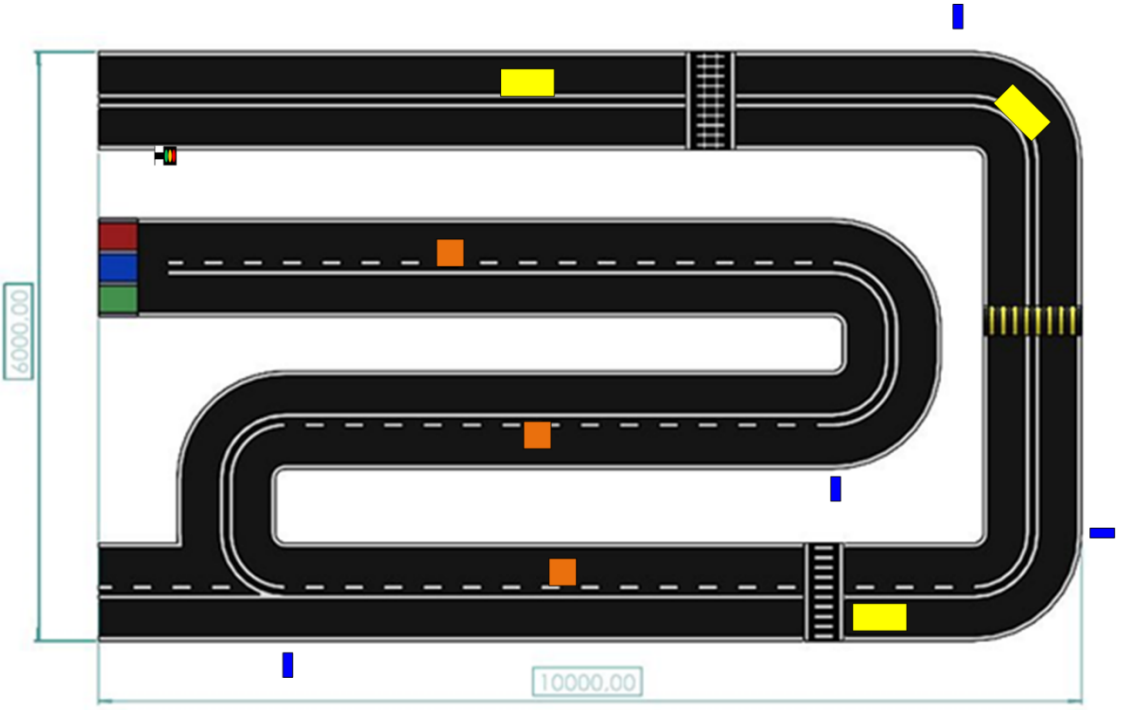


Figure 1. *General Layout of the Vehicle Competition Track and Task Areas*

The competition course consists of a track laid out on a black surface, **as shown in Figure 1**. The course features **solid and dashed road markings**, with the road edges demarcated by **white stripes**. Autonomous vehicles must remain **within their own lanes** throughout the competition.

The track is designed to consist of **a starting area, task areas, overtaking areas, a dead-end section, parking areas**, and a finishing section. The areas where tasks will be performed are positioned **at specific intervals** along the track, and each task is designed to test the competitors' autonomous perception and decision-making abilities.

The zone completion areas on the course are marked by **blue signs** placed at the roadside. Autonomous vehicles must complete these zones **without leaving the course or violating lanes**.

At the starting point of the course, there is a **traffic light system** that controls the start of the race, and the race is started via this system. At the end of the course, there are **coloured parking areas** where autonomous vehicles will perform the parking task.

For **the overtaking** task, **one vehicle** will be used on the track. The position of this vehicle **will be redefined by the referee for each competitor**. The vehicle to be overtaken will only be placed in any of **the areas where overtaking is not prohibited**. Example positions of the vehicle to be overtaken are shown **in Figure 1**. This vehicle is **orange in colour** and designed with dimensions of **20 × 30 × 25 cm**.

In **areas** where **overtaking is prohibited**, **yellow** vehicles measuring **20 × 45 × 25 cm** will be placed at random positions on the track, **as shown in Figure 1**. These vehicles will be used as visual obstacles and traffic elements in the opposite lane.

The detailed measurements, geometric structures, **and placement characteristics** of the task objects and components located on the course are explained under the heading **3.2 Definition of Task Objects and Components**.

3.2. Definition of Task Objects and Components

This section defines the dimensions, shapes, and visual characteristics of the task objects and components used on the competition course. All dimensions and placement details for **road markings, pedestrian crossings, level crossings, speed bumps, dead-end streets, parking areas, traffic lights, and warning signs** are shown in the relevant figures (Figure 2- Figure 10).

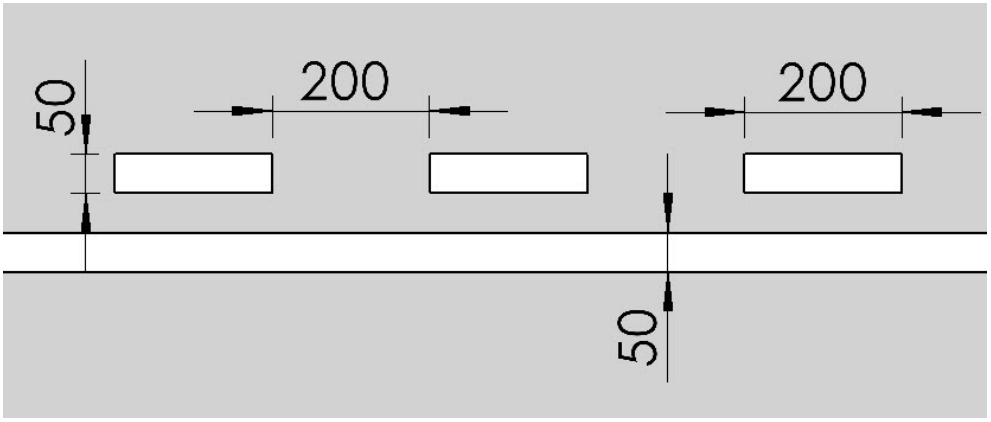


Figure 2. *Road marking dimensions and placement intervals*

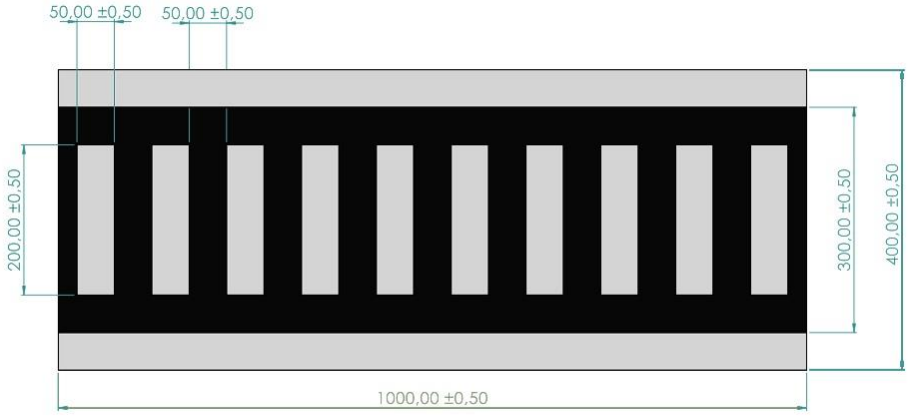


Figure 3. *Pedestrian crossing dimensions and placement intervals*

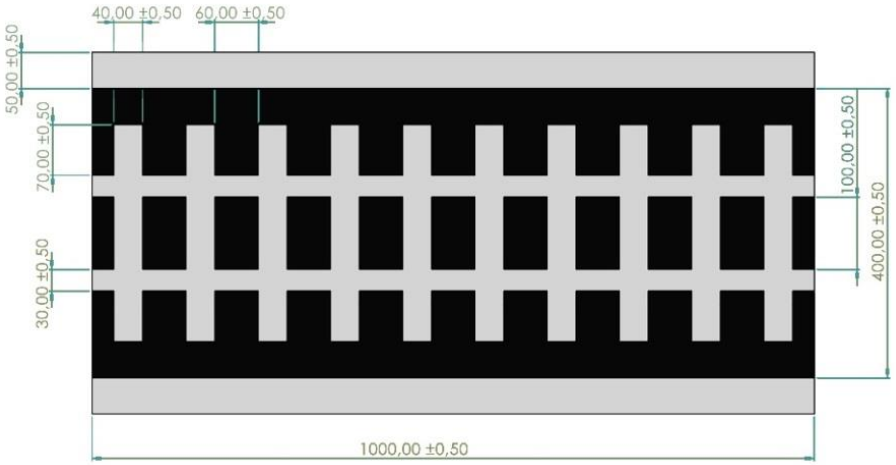


Figure 4. *Level crossing dimensions and placement intervals*

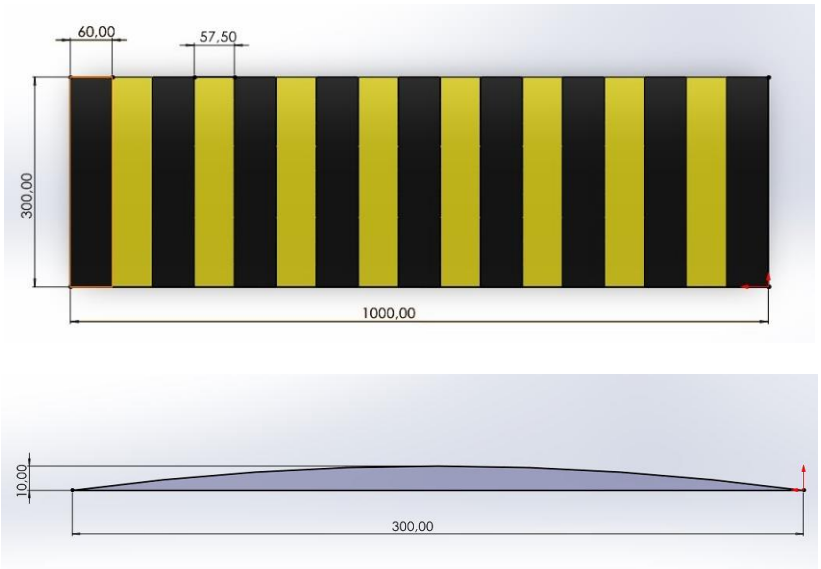


Figure 5. *Speed bump dimensions and placement intervals*

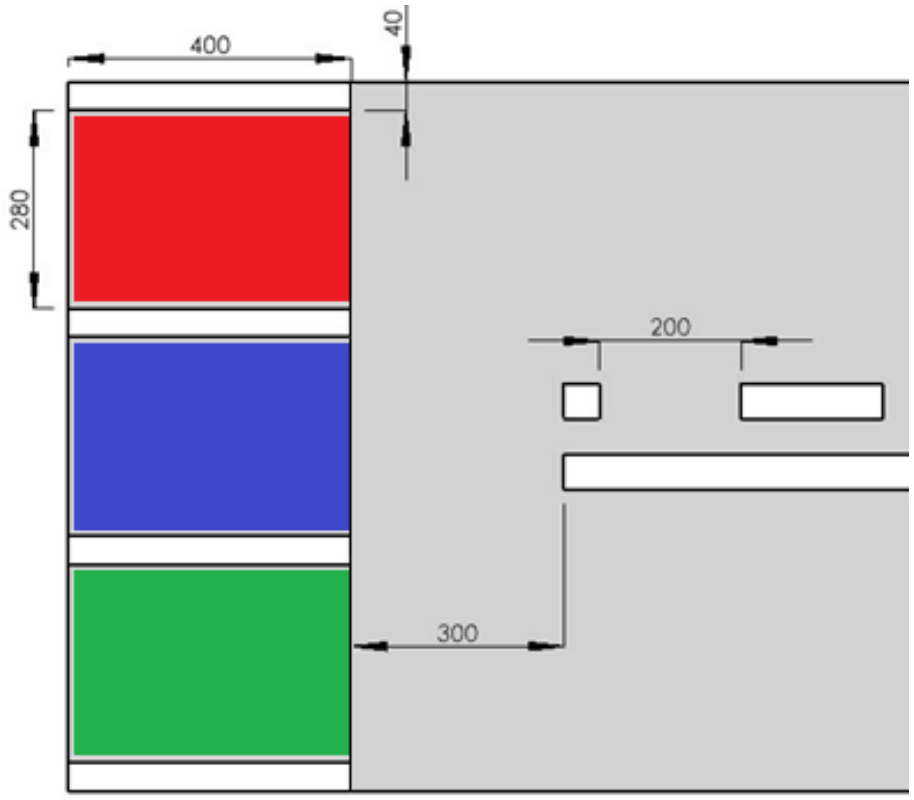


Figure 6. *Parking area dimensions and placement intervals*

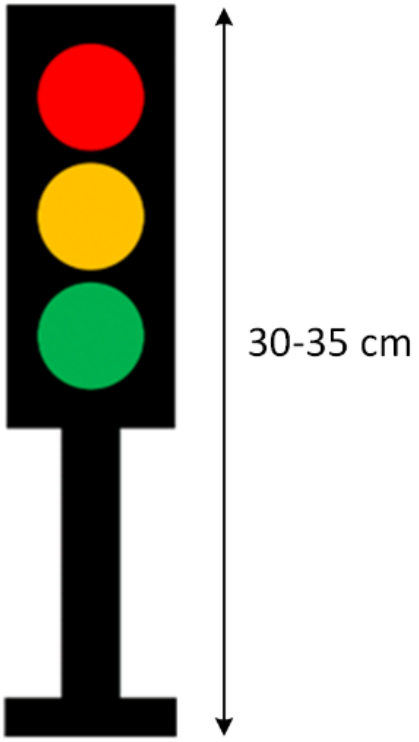


Figure 7. *Traffic Light*

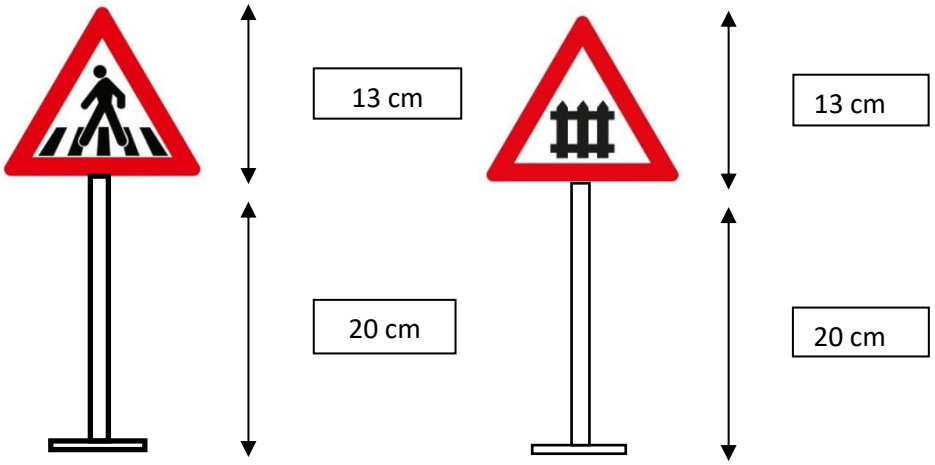


Figure 8. *Dimensions of pedestrian crossing and level crossing warning signs*

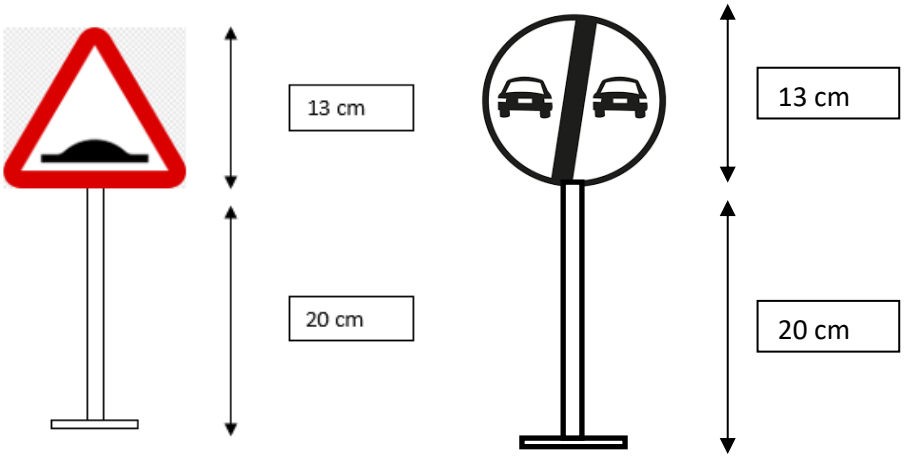


Figure 9. *Dimensions of speed bump and overtaking permitted warning signs*

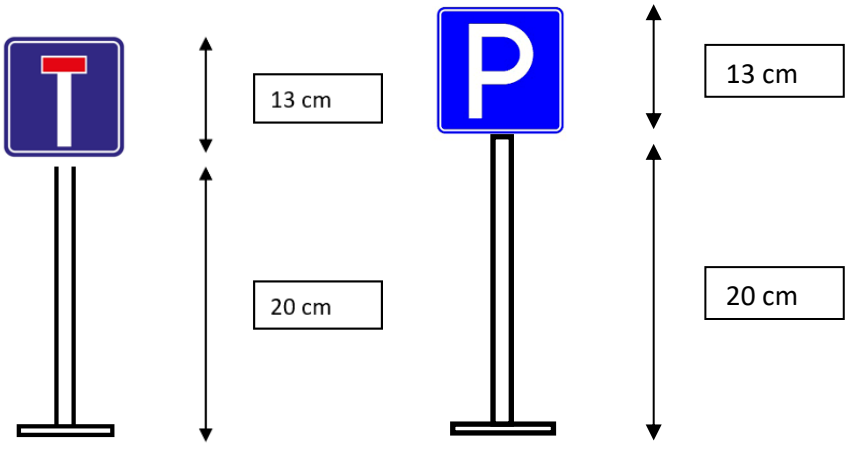


Figure 10. *Dimensions of dead-end road and parking area warning signs*

3.3. Object Placement Procedure and Tolerance

All objects, signs and components located on the competition course shall be positioned according to the reference points defined in this guide. The dimensions and tolerances specified for the positioning of objects shall be adhered to. Any positioning outside these tolerances shall be deemed invalid. Judges shall check the positioning of all objects on the course prior to the competition.

The principles for the placement of objects are outlined below:

- **Traffic Light:**

  The traffic light is placed **1 metre** ahead of the start of the track. **A tolerance of ±10%** applies to this distance.

- **Pedestrian Crossing Sign:**

  The pedestrian crossing sign is placed at the edge of the course, **aligned with the starting point of the pedestrian crossing**.

- **Speed Bump Sign:**

  The speed bump sign is placed at the edge of the track**, aligned with the starting point of the speed bump**.

- **Level Crossing Sign:**

  The level crossing sign is placed at the edge of the track, **aligned with the starting point** of the level crossing.

- **Overtaking Permitted Sign:**

  The overtaking permitted sign is placed at the edge of the track, **aligned with the starting point** of the overtaking permitted zone.

- **Parking Area Sign:**

  The parking area sign is placed at the edge of the track, **aligned with the starting point** of the parking area.

- **Dead End Sign:**

  Dead-end road signs are placed **along the edge** of the track, **aligned with the starting point** of the dead-end road section.

The positions and order of pedestrian crossings, level crossings and speed bumps on the track are determined **by the referees** before the robot begins competing in the relevant round. The locations and order of these objects on the track can be changed for **each competitor**.

3.4. Definition of Duties and Conditions of Implementation

This section explains the definition, implementation method, and evaluation criteria of the tasks on the course within the Autonomous Vehicle Category. Competitors' autonomous vehicles are expected to perform the tasks according to the specified rules and using only camera-based perception.

3.4.1.Task 1: Traffic Light Detection and Start Task

Teams that prepare **the autonomous vehicle** for operation via pre-loaded software, **without requiring an external computer connection**, using **a button or similar trigger element**, will be awarded **50 bonus points**.

The traffic light in the start section will flash **red, amber, and green**. **Upon** the traffic light **turning green**, autonomous vehicles must start moving and begin the race **within a maximum of 3 seconds**. The intervals between the lights are randomly determined.

Autonomous vehicles that fail to start are granted **a technical intervention period** equal to the time specified on the race day, provided they remain within the field. At the end of this period, the autonomous vehicles are asked to start again.

Autonomous vehicles that start correctly in accordance with the rules in the first attempt are awarded **50 bonus points**, while those that start correctly in the second attempt are awarded **25 bonus points**. Autonomous vehicles that fail to start correctly within the specified time limits **are not awarded any starting points**. However, these vehicles **may continue in the race** to perform other tasks on the course **without being disqualified**.

Autonomous vehicles that do not move **within 15 seconds** after the traffic light turns green on the second attempt **are disqualified** from the relevant round.

3.4.2.Task 2: Pedestrian Crossing Task

Autonomous vehicles must **detect the pedestrian crossing** on the course and stop in front of it. The distance between the point where the autonomous vehicle stops and the pedestrian crossing must be **no more than 30 cm**. The vehicle must wait at this point for **at least 5 seconds**.

Autonomous vehicles that fulfil these conditions in accordance with the rules are awarded **50 points**.

3.4.3.Task 3: Speed Bump Task

Autonomous vehicles **must detect** the **speed bump** on the track and **successfully pass over** it. Autonomous vehicles that pass over the speed bump without leaving the track and remain within their own lane are considered to have successfully completed this task.

If an autonomous vehicle fails to pass the speed bump or gets stuck on it, the competitor is permitted to reposition the vehicle on the track **so that it can pass the speed bump**. After this, the autonomous vehicle **may continue** in the race, but in this case, **no reward points will be awarded** for the speed bump task.

Autonomous vehicles that pass the speed bump by completing this task in accordance with the rules are awarded **50 points**.

3.4.4.Task 4: Level Crossing Task

Autonomous vehicles must **detect** the **level** crossing on the track and stop in front of it. The distance between the point where the autonomous vehicle stops and the level crossing must be **no more than 30 cm**. The vehicle must wait at this point for **at least 5 seconds**.

Autonomous vehicles that fulfil these conditions in accordance with the rules are awarded **50 bonus points**.

3.4.5.Task 5: Vehicle Overtaking Task

Autonomous vehicles **must detect the overtaking zone** on the track and **overtake** the vehicle on the track in this zone **in accordance with the rules**. The overtaking manoeuvre must only be performed within the overtaking zone.

Autonomous vehicles that change lanes in areas where overtaking is prohibited, overtake from the wrong area, or fail to complete the overtaking manoeuvre are considered to have **failed** this task.

Autonomous vehicles that complete this task in accordance with the rules are awarded **100 reward points**.

3.4.6.Task 6: Dead-End Road Detection and Direction Change Task

Autonomous vehicles must detect **dead-end road sections or dead-end road signs** on the course and continue correctly on the course by performing a right turn manoeuvre without entering the dead-end road.

Autonomous vehicles that enter the dead-end road section, proceed within the dead-end road, or pass the dead-end road start and turn onto the wrong route are deemed to have failed this task. Robots that enter the dead-end road are placed back in their lane with **the turn completed**, as decided by the judges.

Autonomous vehicles that complete this task in accordance with the rules are awarded **100 reward points**.

3.4.7.Task 7: Parking Task

The parking areas at the end of the course consist of three different colours: **red, blue,** and **green**. The positions of these parking areas on the course **are changed before each race**.

Autonomous vehicles must park **in the red parking area** at the end of the track.

The autonomous vehicle's parking task is considered complete when the vehicle comes to a stop within the red parking area. Autonomous vehicles that park in the wrong colour parking area or extend beyond the parking area boundaries are considered to have **failed** this task.

Autonomous vehicles that complete this task in accordance with the rules are awarded **100 points**.

3.4.8.Task 8: Area Completion Task

There are **multiple zones** on the course representing task areas. Autonomous vehicles must complete these zones **without leaving the course**. **In zones where overtaking is prohibited**, autonomous vehicles must complete the zones **while remaining within their own lanes**.

Autonomous vehicles that complete the zones **without leaving the course and without lane violations** are awarded **50 zone completion points**.

Lane violation **occurs when any part of** the autonomous vehicle **crosses the white lane** dividing **line** and **moves onto the black surface belonging to the opposite lane**.

An autonomous vehicle that **exits the course** shall be placed by the competitor **at the point where it exited, aligned with that point and within its own lane**, and shall continue the race.

If **an autonomous vehicle** that **has committed a lane violation** comes into contact with an oncoming vehicle, the vehicle **is placed back in its own lane by the referee's decision** and is allowed to continue the race.

If the vehicle **continuously leaves the track**, **it will be disqualified** from the relevant lap **by the referee's decision**. Continuous lane violations **are not grounds for disqualification**.

4. COMPETITION FORMAT AND EVALUATION CRITERIA

4.1. Application and Reporting Process

The application process for the Autonomous Vehicle Category, the conditions for participation in the competition, and the submission procedures are carried out as specified **in the General Application Guide**.

No robot production or design report **is** required from competing teams in this category**.** Participation in the competition is granted upon fulfilment of the application requirements specified in the General Application Guide.

4.2. Competition Stages

The Autonomous Vehicle Category competitions consist of trial rounds and ranking rounds. The competition process is conducted in a fair and controlled manner for all teams and is carried out in accordance with the stages outlined below.

4.2.1.Test Rounds

Depending on the number of accepted applicants, autonomous vehicles will be permitted to test on the track. Each team will be given equal time during the testing phase.

4.2.2.Qualifying Round

Once the number of autonomous vehicles is determined, the number of laps each vehicle will race will be established. If an autonomous vehicle races two or more laps, the total points will be calculated by summing the lap points.

4.3. Scoring System and Evaluation

The performance of teams competing in the Autonomous Vehicle Category is evaluated based on **task completion points**, **rule violations**, and **competition duration**. The award points for each task and the task-based evaluation criteria are specified under the heading **3.4. Definition of Tasks and Conditions of Application**.

Task points are awarded only if the conditions defined in the guide for the relevant task are completed **correctly and in full**. If tasks are performed incompletely, incorrectly, or in violation of the rules, **no reward points are awarded** for the relevant task. Penalties for situations such as leaving the course or lane violations are assessed according to the principles specified in the relevant task definitions.

The competition course is considered **one lap**. If more than one lap is completed, **the total score** of the competing teams is calculated by adding the scores obtained in all laps completed. No points are calculated for laps that result in disqualification.

For autonomous vehicles that successfully complete the course, the race time is included as an additional criterion in the evaluation. **The completion time coefficient** is calculated using the following formula:

$$\text{Bitirme Süresi Katsayısı} = (4 \times 60) - \text{Bitirme Süresi (sn)}$$

This coefficient **is added to** the total points earned from the tasks. The completion time coefficient applies only to autonomous vehicles that complete the course.

In the event of a tie, the autonomous vehicle **that completes** the course **in the shortest time** is ranked higher. If the tie persists, the judging panel determines the ranking based on the general evaluation criteria for competition performance.

All points earned and evaluations made during the competition are recorded by the judges, and the final results are announced at the end of the competition.

4.4. Race Duration and Break Usage

In the Autonomous Vehicle Category, **the maximum race time** for each race lap **is 4 minutes (240 seconds)**. The autonomous vehicle must complete the course within this time. For autonomous vehicles that fail to complete the course within the specified maximum time, the relevant lap **is terminated by the referee at , but the points earned from tasks completed in accordance with the rules up to that time are valid**.

During the competition, if the autonomous vehicle **leaves the track** or **gets stuck by failing to pass the speed bump**, **technical intervention** is permitted with the referee's approval. As part of the technical intervention, the vehicle is placed on the track **at the point it left, within its own lane**, or **in a position where it would have passed the speed bump**. In cases where technical intervention is performed, **no reward points are awarded** for the relevant task.

Competitors **are not granted breaks** during each round. However, **breaks for preparation and inspection** may be granted before the competition begins or between rounds, during periods announced by the referees. Any interventions on the track or on the autonomous vehicle outside these periods shall be considered a **rule violation**.

Any intervention on the autonomous vehicle during the competition **without the referee's permission** shall be considered grounds for **disqualification** for the relevant lap. The competition time begins **when** the autonomous vehicle **moves from the starting signal** and ends when the vehicle completes the course or when the lap is terminated by the referee.

5. ETHICAL AND OTHER RULES

5.1. Disqualification and Penalty Situations

- Competitors who fail to comply with the rules specified **in the Application Guide**, engage in unethical behaviour, or **violate the provisions of the technical specifications shall be disqualified**.
- Autonomous vehicles are prohibited from leaving **permanent marks, signs or damage** on the track. Autonomous vehicles found to have damaged the track **will be disqualified**.

- Autonomous vehicles may use **electrical energy sources such as batteries or battery packs**. The use of **liquid, flammable, or explosive energy sources** is prohibited. The use of such sources **is grounds for disqualification**.
- During competitions, **only tyre or battery** changes are permitted on autonomous vehicles. If the body of the autonomous vehicle is modified, the vehicle **will be disqualified**.
- During competitions, if **the QR code** affixed to the autonomous vehicle at the registration desk **is removed, relocated, or damaged**, the autonomous vehicle **will be disqualified**.
- Vehicles that **do not match the photographs of autonomous vehicles** registered in the system at the judges' desk **will be disqualified**.
- If electronic circuit components need to be replaced, components **of the same type and with the same specifications** may be replaced in **the same position**. If the square code is damaged during this process, the autonomous vehicle **will be disqualified**.
- The square code must be affixed to **the fixed body** of the autonomous vehicle. Square codes affixed to **removable parts** are considered invalid. If this is detected, the referees **may disqualify** the autonomous vehicle.
- Minor changes may be made to the track dimensions during construction, **provided they do not alter the overall structure**. Such changes shall not be grounds for protest by competitors.
- During the competitions, any objections made due to **the illuminated screens, scrolling text, cameras, and lighting systems** around the track **shall be deemed invalid**.
- **The Competition Organising Committee** reserves the right to **amend, update or add to** the rules set out in the guide as it deems necessary. Any changes made shall be deemed valid from the moment they are announced.

5.2. Objection Procedure

Teams may lodge protests regarding competition results or referee decisions after the completion of each round.

Objections must be submitted to the competition organisation **within 30 minutes** of the end of the relevant round.

Appeals submitted outside the specified timeframe **will not be considered**. Appeals are reviewed by the competition committee, and the decisions made **are final**.

5.3. Warnings and Ethical Rules for Competitors

The Application Guide must be read carefully before applying for the competition. General rules regarding competition applications and the Autonomous Vehicle Category are outlined **in the Application Guide at** . All teams applying for the competition are expected to **carefully review** the Application Guide before beginning the competition process.

Competitors called to the track on the day of the competition **will not be given additional time** to charge their autonomous vehicles' batteries. Therefore, it is the competitors' responsibility to ensure that their autonomous vehicles **are ready for the competition** before being called to the track.

Competitors are expected to demonstrate **ethical**, fair and sportsmanlike behaviour throughout the competition. Behaviour that disrupts the order of the competition or affects other competitors or judges should be avoided.

Complaints regarding environmental elements such as **illuminated screens, scrolling text, cameras, and lighting systems** located around the track during the competition will not be considered.

Minor changes may be made to the track dimensions and course layout during the construction phase, **provided they do not disrupt the overall structure**. Such changes are considered a natural part of the competition and do not constitute grounds for protest.

The Competition Organising Committee reserves the right to **update or amend** the rules set out in the guide as deemed necessary. Any changes made shall be considered valid from the moment they are announced to the competitors.

5.4. Safety Measures

Category-specific safety measures that must be followed Vehicles competing in the Autonomous Vehicle Category must be designed and used in such a way that **they do not cause harm to people, the environment, or the track infrastructure** during the competition.

In this context;

- The use **of dangerous mechanical devices** in autonomous vehicles that could cause **cutting, piercing, injury**, throwing or impact effects is prohibited.
- The use of **liquid, flammable, explosive or pressurised** energy sources and systems containing such substances is prohibited.
- Autonomous vehicles identified as posing a security risk **may be excluded from the race** by the referee's decision.

5.5. Competition Organising Committee Powers

The Competition Organising Committee **has the authority to revise the format of** the competition as deemed necessary.

These changes are communicated to competitors through official channels and are binding for all teams.

Question and Announcement Tracking Channel

All announcements made during the competition must be followed through the official platforms designated by the participants.

The rankings resulting from the completion of the competition rounds**,** as well as **new competition dates and times**, will be announced via these platforms.

6. CONTACT

6.1. Competition Coordination Information

Communication regarding the competition will be provided through the official email address, telephone number and responsible unit to be announced. Competitors must use only these official communication channels.

7. ATTACHMENTS

7.1. Competition Card

18. ULUSLARARASI MEB ROBOT YARIŞMASI
OTONOM ARAÇ KATEGORİSİ **ELEME TURU**
MÜSABAKA KARTI

ROBOT			MÜSABAKA NO
NO	«BIRINCIYARISMACI»	GELMEDİ O DİSKALİFİYE O Sebebini açıklama alanına yazınız	«MacKodu» ELEME TURU

FOTOĞRAF KONTROL		TARİH SAAT:
FARKLI O	AYNI O	..../05/2026 ---- /-----
Robotu sistemdeki fotoğrafıyla kontrol ederek işaretleyiniz.		

MÜSABAKA PUANLAMASI

ÖDÜL PUANLARI		NOTLAR				TOPLAM
BAŞLANGIÇ ÖDÜL PUANI (Trafik Işığına Göre) (İlk / İkinci Deneme)	50/25					
BUTON İLE BAŞLATMA EK PUANI	50					
YAY GEÇİDİ ÖDÜL PUANI	50					
HEMZEMİN GEÇİT ÖDÜL PUANI	50					
HIZ TÜMSEĞİ ÖDÜL PUANI	50					
ARAÇ SOLLAMA ÖDÜL PUANI	100					
ÇIKMAZ YOL TANIMA ÖDÜL PUANI	100					
KIRMIZI ALANA PARK ETME ÖDÜL PUANI	100					
BÖLGE TAMAMLAMA ÖDÜL PUANI	50					
BİTİRME SÜRESİ KATSAYISI (puan ilavesi) (4*60 - Yarış tamamlama süresi)	Yarış tamamlama süresi					
					GENEL TOPLAM PUAN	

Yarışma ile ilgili özel bir durum oluşmuşsa aşağıya yazınız.			
HAKEMLER (AD SOYAD İMZA)			
KONTROL HAKEMİ	MUSABAKA HAKEMLERİ		MASA HAKEMİ
	1. HAKEM	2. HAKEM	

Bu kart, tur bitiminde OTONOM ARAÇ Baş Hakemine teslim edilecektir.

7.2. Sample Scenario

Task/Game Principles and Sample Scenario:

The primary objective expected of competitors in the Autonomous Vehicle category is to complete all tasks defined on the course **in accordance with the rules**, using **only camera-based image processing**. The tasks are structured to resemble real traffic conditions, and autonomous vehicles are expected to demonstrate **correct decision-making, safe driving, error-free navigation, and time management** skills.

The tasks are listed below:

Start:

The autonomous vehicle must move within 3 seconds of the traffic light turning green, pass the sensor line, and begin the race.

Pedestrian Crossing:

The vehicle must detect the pedestrian crossing, stop at a distance of no more than 30 cm, and wait for at least 5 seconds.

Level Crossing:

When detecting a level crossing, the vehicle must similarly apply the minimum 5-second stop rule.

Speed Bump Task:

The autonomous vehicle must pass the speed bump on the track without leaving the track and remaining within its own lane. Vehicles that cannot pass the speed bump may continue the race by being positioned as if they had passed the bump with the approval of the referee, but in this case, no speed bump award points will be given.

Vehicle Overtaking Task:

In areas where overtaking is not prohibited, vehicles on the track must be overtaken in accordance with the rules. Overtaking manoeuvres must only be performed within the overtaking zone.

Dead-End Road Detection:

The autonomous vehicle must detect the dead-end road section or signs and perform a right-turn manoeuvre without entering the dead-end road.

Parking:

The vehicle must park accurately in the red parking space out of the three-coloured parking spaces (red, blue, green) located at the end of the course. The positions of the parking spaces on the course may be changed before the competition.

Area Completion:

The autonomous vehicle must complete multiple zones on the course without exiting the course. In zones where overtaking is prohibited, the vehicle must remain within its own lane. In the event of a lane violation or leaving the course, the relevant zone completion bonus points will not be awarded.

How Success is Measured:

Success is evaluated based on task completion points, no rule violations, and efficient use of the competition time.

Completing each task in accordance with the conditions defined in the guide ensures that the relevant reward points are earned.

In cases of lane violation or leaving the course, the relevant task's reward points will not be awarded.

A vehicle that leaves the course may continue the race after being placed back in its lane with the referee's approval , but this situation does not earn points.

For autonomous vehicles that successfully complete the course, the completion time coefficient is calculated as follows and added to the total score:

$$\text{Bitirme Süresi Katsayısı} = (4 \times 60) - \text{Bitirme Süresi (sn)}$$

In the event of a tie in points, the autonomous vehicle that completes the course in the shortest time is ranked higher.

Overall Evaluation:

Autonomous vehicles that complete all tasks **in accordance with the rules**, **consistently** and **in the shortest time**; without leaving the course, without lane violations, and parking correctly in the red parking area, achieve the highest success score in the race. The accuracy of the image processing algorithm, decision-making speed, and autonomous driving stability are among the decisive factors in the evaluation process.